

Seminar: Software Engineering in the Age of AI  
University of Haifa — Department of Information Systems

# Clinical Lab Analysis System

*A controlled, self-correcting generated-code pipeline for answering predefined clinical laboratory questions over MIMIC-III*

## Final Report

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# 1. Research Question, Goals and Importance

## 1.1 Research question

How can laboratory test results recorded during a hospital admission be used to automatically and reliably answer a limited, well-defined set of clinical laboratory analysis questions about hospitalized patients?

## 1.2 The engineering problem

Raw MIMIC-III laboratory data is large (close to a million lab-event rows once merged with admissions) and hard to query ad hoc. Clinicians reviewing a single admission and researchers exploring lab patterns across a cohort both need quick, correct answers to a small set of recurring questions, such as which results during a stay were abnormal, what the latest value of a test is, or how a value trended over time. At the same time, the seminar's broader theme — software engineering in the age of AI — raises a second, harder problem: large language models can now write the code needed to answer such questions on demand, but code generated by a model cannot be trusted blindly, especially over medical data, because it may reference the wrong columns, silently return an empty or malformed result, or execute an unsafe operation. The project therefore treats “generate code to answer a question” and “make sure that generated code is correct and safe before trusting its output” as one combined engineering problem, rather than two separate ones.

## 1.3 Why it matters

- Clinical relevance: fast, correct answers about abnormal results, trends, and admission-level summaries support bedside review by a doctor and reproducible, exportable analysis by a researcher, without requiring either user to write SQL or pandas code themselves.
- Software-engineering relevance: the project is a concrete case study in building an auditable “generated code” system — one that shows its work (the code it ran), checks its own output (validation), attempts to repair itself (correction), and only cautiously extends into open-ended LLM code generation behind a sandbox. This is directly relevant to the seminar's subject: how to engineer software responsibly when part of the pipeline is AI-generated.
- Safety in a sensitive domain: healthcare data raises the cost of a wrong or hallucinated answer. Restricting the system to a fixed, validated question registry by default — and only optionally allowing free-form LLM-authored code inside a hardened sandbox — is itself a design decision about how much autonomy to give generated code in a high-stakes domain.

## 1.4 Goals and objectives

1. Support a fixed, precisely specified set of clinical laboratory questions (12 in the final system) over a cleaned MIMIC-III merge of ADMISSIONS and LABEVENTS.
2. For every question, automatically generate executable Python code, run it, validate its output against explicit structural and clinical rules, and apply a bounded, rule-based self-correction step when execution or validation fails.
3. Present the result together with the generated code, the validation outcome, and the correction attempts, so a user can audit how an answer was produced, not just see the answer itself.
4. Offer an optional controlled natural-language interface that either maps user requests to one of the supported question templates or, in the experimental AI mode, invokes the guarded sandboxed code-generation pipeline.

5. As a bounded, off-by-default experiment, evaluate whether a guarded, sandboxed LLM “programmer agent” loop (inspired by the concepts presented in AgentCoder and Self-Edit, see section 7) can safely extend the system to free-form questions without weakening the guarantees of the trusted default path.
6. To support different analytical workflows, the final system provides separate Doctor and Researcher workspaces that share the same trusted backend pipeline while exposing questions most relevant to each user role.

## 1.5 Expected contribution

The expected contribution is a small but complete reference implementation of a controlled "generate → execute → validate → correct → present" pipeline for automated clinical laboratory analysis. The system combines a single-source-of-truth question registry that keeps the user interface, code generator, validator, and execution pipeline synchronized; a deterministic validation and rule-based correction layer that detects and repairs a meaningful class of execution errors without requiring additional model calls; and a clearly separated, sandboxed experimental extension that demonstrates how the same engineering principles can be applied to LLM-generated code while maintaining safety through guardrails such as AST allow-listing, restricted builtins, execution timeouts, row limits, and deterministic output validation.

# 2. Engineering Background and Proposed Method

## 2.1 Proposed approach

The system is a layered, registry-driven pipeline. A user selects one of the supported clinical questions (or types a natural-language question that is routed to one); the system generates a short, readable Python snippet that calls a known data-preparation function with the user's parameters; it executes that snippet; it validates the result against structural rules (is it a DataFrame? does it have the expected columns? is it non-empty?) plus a per-question clinical rule (for example, “all rows must be flagged abnormal”); and, only if execution or validation fails, it applies a small rule-based correction and retries, up to a bounded number of attempts. The final result, the generated code, the validation message, and the full correction history are all shown to the user.

```
question id -> generate code -> execute -> validate -> correct (if needed) -> present
```

This constrained pipeline is the trusted default. A second, explicitly optional and off-by-default mode (“Advanced: AI writes code”) provides an alternative execution path that bypasses the fixed registry and invokes an LLM “programmer agent” that writes a pandas snippet for a free-form question; that snippet never runs directly — it is checked against an AST allow-list and then executed inside a restricted sandbox, **following the same generate → execute → validate → retry pipeline before anything is shown to the user.**

## 2.2 Architecture

The codebase is organized into single-responsibility modules, all driven by one configuration object: the question registry.

| Module       | Responsibility                                                           |
|--------------|--------------------------------------------------------------------------|
| data_prep.py | Loads the cleaned dataset once; one data function per supported question |

| Module                        | Responsibility                                                                                                                                                               |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| questions.py                  | QuestionSpec registry (QUESTION_REGISTRY) — the single source of truth: id, label, parameters, data function, expected columns, category, optional validator and chart spec  |
| question_router.py            | Controlled natural-language routing: maps free text to one supported question id + parameters (rule-based; never generates code)                                             |
| llm_client.py / llm_router.py | Optional LLM-assisted routing fallback (Anthropic / OpenAI / Gemini, chosen by which API key is set); the model may only choose a template id + parameters, never write code |
| code_generation.py            | Builds a runnable code string, e.g. <code>result = get_lab_trend(107521, 51221)</code> , from a QuestionSpec + parameters                                                    |
| execution.py                  | Executes the generated code for the registry path and captures success / result / error / status                                                                             |
| validation.py                 | Checks the result is a non-empty DataFrame with the expected columns, plus any per-question rule                                                                             |
| correction.py                 | Bounded, rule-based retry loop: fixes a known set of function-name typos and re-runs                                                                                         |
| result_presentation.py        | Console presentation of a pipeline run                                                                                                                                       |
| main_pipeline.py              | Wires generation → execution → validation → correction → presentation into <code>run_pipeline(question_id, params)</code>                                                    |
| app.py                        | Streamlit UI: role-based Doctor / Researcher workspace, question selection, dynamic parameter widgets, results, tabs                                                         |
| sandbox.py                    | AST allow-list + restricted-builtins execution environment for the optional LLM code-generation path                                                                         |
| code_agent.py                 | The LLM “programmer agent”: builds a schema-aware prompt and asks the model for a pandas snippet                                                                             |
| agent_pipeline.py             | Guarded code-gen loop: programmer agent → sandbox → validator → refine, mirroring correction.py's shape                                                                      |
| presentation_agent.py         | Suggests a chart for a code-gen result (rule-based, optional LLM override)                                                                                                   |

**The Question Registry acts as the central configuration layer of the system. All major components—including the user interface, code generation, validation, and execution pipeline—derive their behavior from the same registry definition, ensuring consistency across the entire application.**

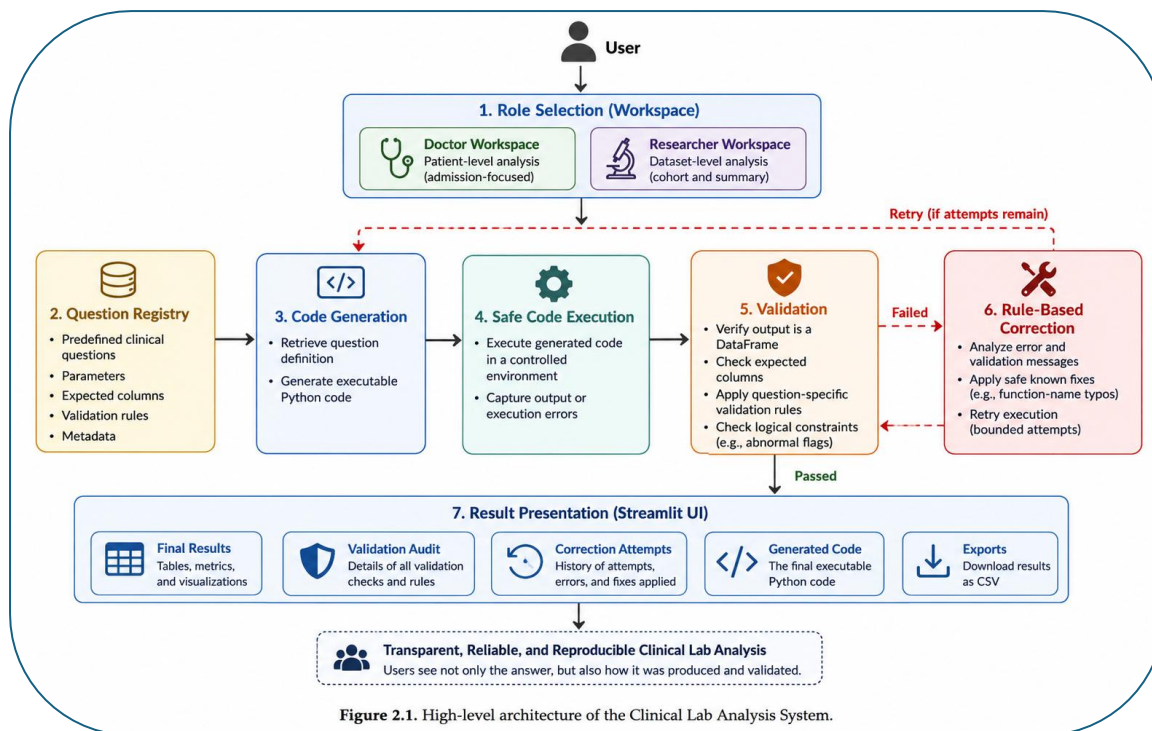


Figure 2.1. High-level architecture of the Clinical Lab Analysis System.

Figure 2.1. High-level architecture of the default, registry-driven pipeline (steps 1-7). “Safe Code Execution” here denotes the guarded try/except wrapper around `exec()` in `execution.py` for the trusted 12-question path; the further-hardened, AST-sandboxed executor (`sandbox.py`) that enforces an allow-list is used specifically by the optional Advanced “AI writes code” path described in section 2.3 and section 3.3.

## 2.3 Algorithms

### Code generation

`generate_code(spec, params)` reads the target data function's own signature with Python's `inspect` module, so it needs no per-question hard-coded argument list. It builds a call such as `result = get_latest_lab_value(199884, 50902)` directly from the `QuestionSpec` and the supplied parameters.

### Execution

For the registry path, `execute_generated_code` runs the snippet with Python's `exec()` against a namespace built only from the registry's known functions (never against arbitrary user code), inside a try/except that captures success, the result object, and any error message and status string.

### Validation

`validate_result(spec, execution_result)` first verifies that execution completed successfully and returned a result. A missing result, an empty string, or an unsupported result type fails validation. A non-empty string is accepted as a valid textual result. For tabular results, the function verifies that the returned value is a pandas DataFrame containing all columns listed in the question's `expected_columns` and rejects an empty DataFrame. The Streamlit presentation layer handles this specific empty-result case separately and displays it as a neutral “no matching records” state. Finally, the validator applies the question-specific `extra_validate` rule, where one exists. Examples include verifying that abnormal-result questions return only `FLAG = abnormal` rows, that “latest value” and summary questions return exactly one row, that trend and date-range questions are sorted by `CHARTTIME` in ascending order, and that dataset-wide aggregate questions satisfy **Abnormal + Normal = Total Tests** for every admission.

## Correction

`run_with_correction` executes and validates the code, and on failure calls `correct_generated_code`, which rewrites a small, explicit map of known function-name typos — matched on whole identifiers only, via regex word boundaries, so a wrong name that happens to be a prefix of a valid one is not corrupted — and retries, up to a bounded `max_attempts` (2 by default). Every attempt — its code, execution result, and validation result — is recorded and shown in the UI's Correction Attempts tab.

## Sandboxed LLM code generation (optional path)

`check_code()` parses a candidate snippet with Python's `ast` module and rejects it unless every node type is included in an explicit allow-list. Imports, loops, function and class definitions, private or dunder attribute access, dangerous builtins, and file-, network-, or evaluation-related methods such as `to_csv`, `read_sql`, `eval`, and `query` are rejected. Accepted code is executed by `run_sandboxed()` inside a worker thread in a restricted namespace containing only `pd`, the provided DataFrame `df`, and a small set of safe builtins. Execution is subject to a soft timeout, and DataFrame or Series outputs are capped to a configured maximum number of rows. The result then passes through the generic `validate_freeform_result()` gate. If execution or validation fails, the error feedback is returned to the LLM for another attempt, up to a bounded attempt limit.

## 2.4 Tools and technologies

- Python 3.10+ (developed and tested on Python 3.13), with pandas for data loading, preprocessing, and analysis, and Streamlit for the interactive web-based user interface, configured with a dark theme via `.streamlit/config.toml`.
- Automated testing using `pytest` and `pytest-cov`, with an 85% minimum code coverage threshold. The final implementation achieved approximately 94% code coverage.
- Optional LLM providers — Anthropic, OpenAI, or Google Gemini — selected according to the available API key in the `.env` file and accessed through a lightweight `httpx`-based client (`llm_client.py`) with automatic retries and a mock/none fallback, allowing the system to operate fully without external API calls.
- Dataset: The MIMIC-III clinical database, using the `ADMISSIONS`, `LABEVENTS`, and `D_LABITEMS` tables, cleaned and merged into a single working CSV file (see Section 3).

## 3. Implementation and Evaluation of the Solution

### 3.1 Inputs

The system is built on three MIMIC-III source tables: `ADMISSIONS` (one row per hospital admission, containing `SUBJECT_ID`, `HADM_ID`, `ADMITTIME`, `DISCHTIME`, and `DIAGNOSIS`), `LABEVENTS` (one row per laboratory measurement, containing `SUBJECT_ID`, `HADM_ID`, `ITEMID`, `CHARTTIME`, `VALUENUM`, `VALUEUOM`, and `FLAG`), and `D_LABITEMS` (a lookup table mapping `ITEMID` values to human-readable laboratory test names). During preprocessing, these tables are cleaned and merged into a single working dataset (`cleaned_merged_dataset.csv`), which is used by the application during runtime.

At runtime, the user selects one of the supported clinical questions and provides the required parameters (a subset of admission ID, laboratory test/ITEMID, patient ID, or date range), depending on the selected question. Alternatively, the user may enter a free-text natural-language question, which is routed to the appropriate predefined question through the controlled routing mechanism.

## 3.2 Data preprocessing

An initial schema-inspection pass (Google Colab, pandas) checked every field the supported questions rely on and quantified missing values in each table:

| Table      | Column                                    | Missing values                |
|------------|-------------------------------------------|-------------------------------|
| ADMISSIONS | SUBJECT_ID, HADM_ID, ADMITTIME, DISCHTIME | 0                             |
| ADMISSIONS | DIAGNOSIS                                 | 25 (non-critical for the MVP) |
| LABEVENTS  | SUBJECT_ID / ITEMID / CHARTTIME           | 1 each                        |
| LABEVENTS  | HADM_ID                                   | 371,649                       |
| LABEVENTS  | VALUENUM                                  | 179,945                       |
| LABEVENTS  | VALUEUOM                                  | 190,809                       |
| LABEVENTS  | FLAG                                      | 994,847                       |
| D_LABITEMS | ITEMID, LABEL                             | 0                             |

Because the system joins LABEVENTS to ADMISSIONS on HADM\_ID, rows with a missing HADM\_ID cannot be attributed to an admission and are dropped. Rows with a missing VALUENUM are excluded only for the numeric analyses that require it (trend, latest value, first-vs-last, summary statistics); other questions (e.g. “all lab tests performed”) do not require it. FLAG has a large number of missing values because most lab results are normal and are simply left unflagged rather than explicitly marked “normal”; this is treated as expected, not as data corruption. The cleaning pipeline (1) keeps only the columns the project needs, (2) removes LABEVENTS rows with a null HADM\_ID, (3) joins LABEVENTS to ADMISSIONS on HADM\_ID, and (4) attaches a readable lab name from D\_LABITEMS via ITEMID. The result, `cleaned_merged_dataset.csv`, contains 1,003,949 rows covering 2,839 admissions, 2,234 patients, and 376 distinct lab test types, with columns SUBJECT\_ID\_x, HADM\_ID, ITEMID, CHARTTIME, VALUENUM, VALUEUOM, FLAG, SUBJECT\_ID\_y, ADMITTIME, DISCHTIME, DIAGNOSIS, LABEL. The file is kept out of version control because of its size and is fetched automatically by the app on first run (or placed manually next to `app.py`).

## 3.3 Main components

The trusted default path is the 12-question registry described in section 2, exposed through a role-based Streamlit UI: a Welcome screen lets the user enter a Doctor workspace (admission-level review — abnormal results, latest value, trend, first-vs-last comparison) or a Researcher workspace (dataset-level exploration — summary statistics, cohort counts, date-window filtering, CSV export), with questions relevant to both roles shown in both. Every question run produces a results area with summary metric cards, an optional trend chart, a colour-coded status (pass / no-records / fail), and four tabs: Final Result (with a CSV download button), Validation Audit, Correction Attempts, and Generated Code. A controlled natural-language box sits alongside the dropdown and is answered by the same registry through `question_router.py`, with an optional

LLM-assisted fallback. The optional Advanced code-generation path provides an experimental execution path in which an LLM generates a pandas snippet for questions outside the fixed registry.

The screenshot shows the 'Clinical Lab Analysis System' interface. At the top, a description states: 'This system takes one predefined clinical laboratory question, generates executable Python code, runs it on the cleaned MIMIC-III laboratory dataset, validates the output, applies correction when needed, and presents the final result. It currently supports 11 question types, all defined in a single question registry.' Below this is a text input field with the placeholder 'Ask a clinical lab question in natural language (optional)'. The 'Question Selection' section has a 'Category' dropdown set to 'All' and a 'Choose a clinical question' dropdown set to 'Show abnormal lab results for a specific admission'. A note below says 'Lists every abnormal flagged lab result for the admission. Inputs: admission.' The 'Admission ID (HADM\_ID)' dropdown is set to '145834.0'. A red 'Run Analysis' button is to the right. The 'Selected Question' section shows '[1] Show abnormal lab results for a specific admission'. The 'Selected Parameters' section shows a JSON object: 

```
{  "hadm_id": 145834}
```

. The 'Admission Context' section contains a table:

| HADM_ID | ADMITTIME           | DISCHTIME           | DIAGNOSIS   |
|---------|---------------------|---------------------|-------------|
| 145834  | 2101-10-20 19:08:00 | 2101-10-31 13:58:00 | HYPOTENSION |

Below the context is a green bar indicating 'Analysis completed successfully'. At the bottom, four status boxes are shown: 'Execution' (Execution completed successfully), 'Validation' (Passed), 'Correction Applied' (No), and 'Overall Status' (Execution and validation completed successfully).

Figure 1. Question 1 (“Show abnormal lab results for a specific admission”) in the Streamlit UI, with the Final Result, Validation Audit, Correction Attempts and Generated Code tabs.

The screenshot shows the 'Clinical Lab Analysis System' interface for a different question. The description is the same. The 'Question Selection' section has the 'Choose a clinical question' dropdown set to 'Show the trend of a selected lab test over time during an admission'. A note below says 'Shows how one lab test changes over time. Inputs: admission, lab test.' The 'Admission ID (HADM\_ID)' dropdown is set to '145834.0'. The 'Lab Test' dropdown is set to 'Alanine Aminotransferase (ALT) | ITEMID: 50861'. A red 'Run Analysis' button is to the right. The 'Selected Question' section shows '[3] Show the trend of a selected lab test over time during an admission'. The 'Selected Parameters' section shows a JSON object: 

```
{  "hadm_id": 145834,  "itemid": 50861}
```

. The 'Admission Context' section contains the same table as in Figure 1. The 'Selected Lab Test' section shows 'Alanine Aminotransferase (ALT) | ITEMID: 50861'. Below the context is a green bar indicating 'Analysis completed successfully'. At the bottom, four status boxes are shown: 'Execution' (Execution completed successfully), 'Validation' (Passed), 'Correction Applied' (No), and 'Overall Status' (Execution and validation completed successfully).

Figure 2. Question 3 (lab-value trend over an admission) rendered as a time-ordered table with an accompanying trend chart.



## 3.4 Results

### Manual evaluation across multiple admissions and patients

To verify the correctness of the implemented questions, each supported question type was manually evaluated using multiple admissions and patients from the cleaned MIMIC-III dataset. The Table presents representative examples of these evaluations:

| # | Example                 | Result                                      |
|---|-------------------------|---------------------------------------------|
| 1 | HADM 145834             | Abnormal results returned (FLAG = abnormal) |
| 2 | HADM 199884, Chloride   | Latest value 103 mEq/L                      |
| 3 | HADM 107521, Hematocrit | Decreasing trend, 57.9% → 44.4%             |
| 4 | HADM 158675             | All lab tests for the admission listed      |
| 5 | HADM 177047, pO2        | First 82, last 103, difference +21 mm Hg    |

### Automated tests

159 automated tests pass with 94% line coverage (an 85% gate is enforced in pyproject.toml), covering every module plus an end-to-end test that drives the Streamlit UI (via AppTest) through all 12 questions, natural-language routing, and the experimental code-generation mode.

| Module                                                                                                         |  | Coverage |
|----------------------------------------------------------------------------------------------------------------|--|----------|
| agent_pipeline.py / code_generation.py / correction.py / execution.py / llm_router.py / result_presentation.py |  | 100%     |
| sandbox.py                                                                                                     |  | 99%      |
| validation.py / data_prep.py                                                                                   |  | 95%      |
| code_agent.py                                                                                                  |  | 94%      |
| app.py / main_pipeline.py                                                                                      |  | 93%      |
| presentation_agent.py                                                                                          |  | 92%      |
| questions.py / question_router.py                                                                              |  | 91%      |
| llm_client.py                                                                                                  |  | 87%      |
| Total                                                                                                          |  | 94%      |

| # | Question             | Expected result                   | Status |
|---|----------------------|-----------------------------------|--------|
| 1 | Abnormal lab results | Table, all FLAG = abnormal        | Pass   |
| 2 | Latest lab value     | Exactly 1 row                     | Pass   |
| 3 | Lab trend over time  | Time-sorted table + line chart    | Pass   |
| 4 | All lab tests        | Table of all tests                | Pass   |
| 5 | First vs last value  | 1-row comparison incl. difference | Pass   |

| Module |                                    |                                            | Coverage |
|--------|------------------------------------|--------------------------------------------|----------|
| 6      | Summary statistics                 | 1 row: count / min / max / mean            | Pass     |
| 7      | Abnormal vs normal counts          | 1 row: total / abnormal / normal           | Pass     |
| 8      | Abnormal results for a lab         | Abnormal rows, or graceful “no records”    | Pass     |
| 9      | Distinct abnormal tests            | Table of tests with abnormal counts        | Pass     |
| 10     | Values in a date range             | Time-sorted table + line chart             | Pass     |
| 11     | All admissions for a patient       | Table of the patient's admissions          | Pass     |
| 12     | Abnormal vs normal, all admissions | One row per admission, most abnormal first | Pass     |

**All twelve predefined clinical laboratory question types successfully passed both manual verification and automated validation, demonstrating that the registry-driven pipeline operates correctly across all supported analysis scenarios.**

Deploy

## Clinical Lab Analysis System

MIMIC-III based system for predefined clinical laboratory questions

This system takes one predefined clinical laboratory question, generates executable Python code, runs it on the cleaned MIMIC-III laboratory dataset, validates the output, applies correction when needed, and presents the final result. It currently supports 11 question types, all defined in a single question registry.

Ask a clinical lab question in natural language (optional)

### Question Selection

Category: All

Choose a clinical question: List all admissions for a specific patient

All admissions recorded for one patient. Inputs: patient.

Patient ID (SUBJECT\_ID): 2.0

Run Analysis

#### Selected Question

[11] List all admissions for a specific patient

#### Selected Parameters

```
{
  "subject_id": 2
}
```

#### Patient Context

Patient 2 has 1 admission(s) in the dataset.

Analysis completed successfully

Execution

Execution completed successfully

Validation

Passed

Correction Applied

No

Overall Status

Execution and validation completed successfully

Figure 3. Question 11 (all admissions for a patient) — one of the end-to-end-tested and browser-verified questions.

Deploy

# Clinical Lab Analysis System

MIMIC-III based system for predefined clinical laboratory questions

This system takes one predefined clinical laboratory question, generates executable Python code, runs it on the cleaned MIMIC-III laboratory dataset, validates the output, applies correction when needed, and presents the final result. It currently supports 11 question types, all defined in a single question registry.

Ask a clinical lab question in natural language (optional)

Your question is routed to one of the supported question templates - no free-form code is generated or run. Examples:

- Show abnormal lab results for admission 145834
- What is the latest chloride value for admission 1998847
- Show hematocrit trend for admission 107521
- Compare first and last pO2 values for admission 177047

Your question

Show hematocrit trend for admission 107521

Interpret & Run

Question Selection

Category

Choose a clinical question

All

Show abnormal lab results for a specific admission

Lists every abnormal-labeled lab result for the admission, inputs: admission.

Admission ID (HADM\_ID)

145834.0

Run Analysis

Selected Question

[3] Show the trend of a selected lab test over time during an admission

Selected Parameters

```
{
  "hadm_id": 107521
  "itemid": 51221
}
```

Admission Context

| HADM_ID | ADMITTIME           | DISCHTIME           | DIAGNOSIS |
|---------|---------------------|---------------------|-----------|
| 107521  | 2158-04-05 08:54:00 | 2158-04-09 13:34:00 | NEWBORN   |

Deploy

Lists every abnormal-flagged lab result for the admission. Inputs: admission.

Admission ID (HADM\_ID)  
145834.0

Run Analysis

Selected Question

[3] Show the trend of a selected lab test over time during an admission

Selected Parameters

```
{
  "hadm_id" : 107521
  "item_id" : 51221
}
```

Admission Context

| HADM_ID | ADMITTIME           | DISCHTIME           | DIAGNOSIS |
|---------|---------------------|---------------------|-----------|
| 107521  | 2158-04-05 08:54:00 | 2158-04-09 13:34:00 | NEWBORN   |

Interpreted as: [3] Show the trend of a selected lab test over time during an admission | admission 107521, lab Hematocrit (ITEMID 51221)

Analysis completed successfully

Execution  
Execution completed successfully

Validation  
Passed

Correction Applied  
No

Overall Status  
Execution and validation completed successfully

Final Result

Generated Code

Validation Details

Correction Attempts

Validation Details

Validation passed successfully.

Overall status:  
Execution and validation completed successfully

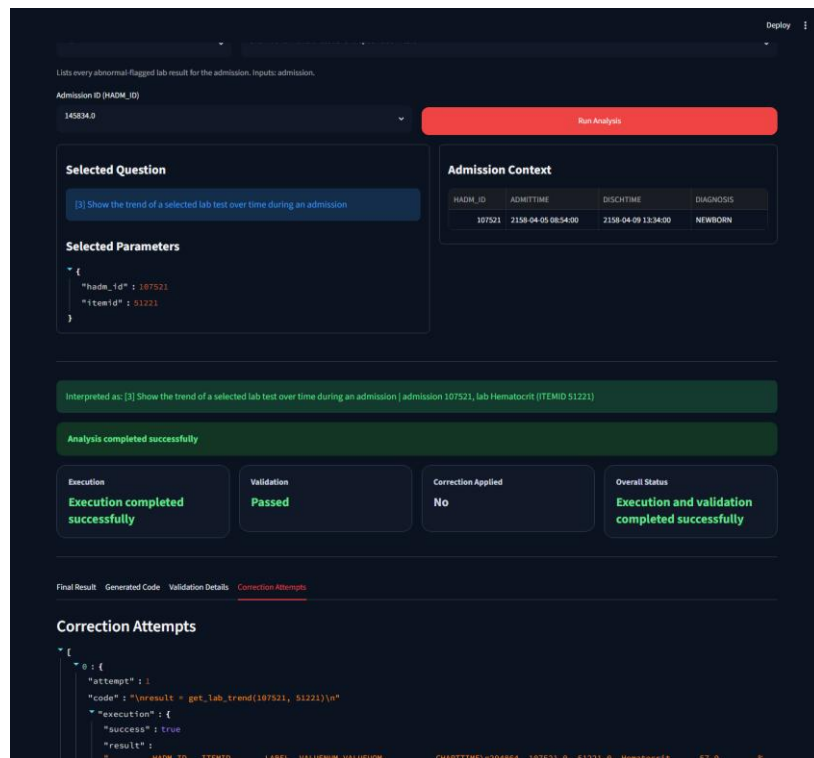


Figure 6. The Correction Attempts tab after the “inject a typo” demo: attempt 1 fails on a misspelled function name and is auto-repaired for attempt 2.

## Edge cases exercised

- No-records path: a question with no matching data shows a neutral “no matching records” state rather than an error.
- Correction loop: a deliberately misspelled function name is auto-corrected and re-run (used as a live demo, since the real generator never emits a typo on its own).
- Regression guard: the typo corrector no longer corrupts the valid name `get_abnormal_results_for_lab`, which contains a shorter valid identifier as a substring.
- Validation failures: execution errors, a `None` result, a non-DataFrame result, missing columns, an empty frame, and a failing per-question rule are all handled and reported distinctly.
- Dataset-wide aggregate (Q12): per-admission counts are cross-checked against the single-admission view (Q7), and `Abnormal + Normal = Total Tests` is validated for every row.
- Code-gen sandbox: imports, open, dunder/underscore attributes, file/network/eval methods, loops, and syntax errors are all rejected by the AST allow-list; runtime errors, timeouts, and row caps are exercised with a fake LLM that is made to fail, be blocked, and refine.

## 3.5 Comparison to baselines and discussion

The project does not have an external, pre-existing system to benchmark against, so three internal baselines were used to justify each guardrail rather than assuming it a priori.

- **Unguarded execution as a baseline.** Running the same generated snippet with a plain `exec()` and no validation or correction step silently returns whatever the code produced—including an empty DataFrame or an output with an incorrect structure—with no indication to the user that the result may be invalid. Adding the validation layer converts such silent failures into explicit validation outcomes (for example, missing columns, empty results, or failed clinical rules), while the correction

layer automatically recovers a meaningful class of failures (such as function-name typos) instead of exposing them directly to the user.

- **Rule-based routing as a baseline for natural-language input.** The default router (`question_router.route_question`) uses deterministic keyword and pattern matching. Comparing it with the optional LLM-assisted fallback (`llm_router.py`) demonstrates the expected trade-off: the rule-based router is fully reproducible and requires no external API calls, but it recognizes only phrasings covered by its predefined rules. The optional LLM fallback (used only when the rule-based routing fails and an API key is available) understands a wider range of natural-language formulations at the cost of external dependency and reduced determinism. For this reason, it is implemented as a fallback rather than the default routing mechanism.
- **The trusted registry path as a baseline for the experimental AI-writes-code path.** Across the 12 supported registry questions, the deterministic pipeline demonstrated consistent and reliable behaviour throughout the evaluation. In addition, the automated test suite (159 passing tests) verified the correctness of both the individual modules and the complete end-to-end pipeline. The optional Advanced mode trades some of this predictability for flexibility: it can answer questions outside the fixed registry, but its AST allow-list intentionally rejects unsupported language constructs (such as loops and imports), and its output is validated using the more general `validate_freeform_result()` function, which verifies the result's type and structure rather than applying question-specific clinical rules. This comparison supports the decision to keep the registry-driven pipeline as the trusted default and the AI-generated code path as an explicitly labelled, opt-in experimental extension.

**Overall, the evaluation supports the project's central claim: a small, fully specified set of clinical laboratory questions can be answered reliably through a generate → execute → validate → correct pipeline, while more open-ended LLM-generated analyses can be supported only when execution is sandboxed and every result is still verified by a deterministic validation stage before being presented to the user.**

## 4. Development Process, Challenges and Lessons

### 4.1 Development phases

1. Steps 1–4 — Scope and data: froze the MVP scope (a small predefined question set only, ADMISSIONS + LABEVENTS + optional D\_LABITEMS, lab rows restricted to a valid HADM\_ID), chose the first 5 question types, inspected the raw MIMIC-III schema in Google Colab (missing-value counts per column, see section 3.2), and built the cleaned, filtered, merged working dataset.
2. Steps 5–16 — First MVP: defined the 5-module architecture (Question Selection, Data Preparation, Code Generation, Validation and Correction, Result Presentation), picked the Python + pandas + Streamlit stack, and implemented each module — first against mock/sample DataFrames to validate the interfaces, then wired to the real dataset — arriving at a first working end-to-end console pipeline before building the Streamlit UI.
3. Step 17 — Registry refactor and polish: replaced the 5 hard-coded questions with a single config-driven QUESTION\_REGISTRY (`questions.py`) that every other module reads from, added 6 further question types (11 total), and polished the UI — a category filter, per-question dynamic parameter widgets, a trend chart, a status banner that reflects success / no-records / failure, and a pinned dark theme.

4. Step 18 and follow-ups — Final deliverables and extensions: wrote the project report, slide deck, and demo script; added a 12th, param-less, dataset-wide aggregate question; added controlled (then optionally LLM-assisted) natural-language routing; added the experimental, sandboxed “Advanced: AI writes code” agent loop inspired by concepts presented in AgentCoder and Self-Edit; and added the role-based Doctor / Researcher workspace UI.

## 4.2 Challenges and how they were resolved

### The cleaned dataset was too large for git

cleaned\_merged\_dataset.csv (~130 MB) exceeds GitHub's 100 MB per-file limit, and Git LFS was judged not worth its quota/setup overhead for a course project. Resolution: the dataset is kept out of version control (.gitignore) and the app downloads it automatically on first run from a shared location, with a documented manual fallback.

### A naive typo-correction rule corrupted a valid function name

The first version of the rule-based corrector used plain substring replacement, so the fix for the typo `get_abnormal_result` silently rewrote the valid, longer identifier `get_abnormal_results_for_lab` (which contains that substring) into something else. Resolution: the corrector now matches on whole identifiers only, using regex word boundaries, and a regression test pins this behaviour.

### Theme inconsistency made the UI unreadable for some users

Headers and cards were unreadable when a viewer's system theme did not match the app's assumptions. Resolution: the Streamlit theme was pinned explicitly in `.streamlit/config.toml` so the app renders identically regardless of the viewer's OS/browser theme.

### How much autonomy to give generated code

The team's initial scope decision was to exclude any LLM code generation from the MVP, since free-text/unbounded questions would break the validator's assumption of known expected columns and could execute arbitrary code. This decision was later revisited, but only by adding a second, clearly optional and off-by-default path, gated by a purpose-built sandbox (AST allow-list, restricted builtins, worker-thread timeout, row cap) built before any LLM-authored code was allowed to run at all — rather than relaxing the guarantees of the trusted registry path.

### Keeping a growing question set consistent across modules

As the question count grew from 5 toward 12, keeping the UI, code generator, validator, and pipeline in sync by hand became error-prone. Resolution: the registry refactor introduced `questions.py` as the single source of truth for all supported questions. The user interface, code generation, validation, and execution pipeline all derive their behaviour from this registry, so adding a new question now requires updating only one central definition.

## 4.3 Lessons learned

- A single source of truth (the question registry) is worth the refactor: it removes an entire class of drift bugs between the UI, the generator, and the validator, and it made every later addition (questions 6–12, the role-based UI, the code-gen agent) a localized change.

- Deterministic validation plus a bounded correction loop is a cheap, high-value safety net — it turns silent or malformed output into an explicit, auditable status without needing a model call for the common failure modes.
- Sandboxing has to come before capability: the AST allow-list, restricted builtins, timeout, and row cap were designed and tested before any LLM-authored code was allowed to execute, not retrofitted afterward.
- Lightweight, in-repository orchestration was preferable to adopting a heavier multi-agent framework (such as CrewAI, AutoGen, or LangGraph) for the experimental code-generation loop. This decision kept the implementation lightweight, easier to debug, and more appropriate for the project's educational scope. The overall design is also consistent with the ideas presented in AgentCoder, where clearly separated responsibilities are emphasized without requiring complex orchestration frameworks.

## 5. Summary and Future Work

### 5.1 Conclusions

The project demonstrates that a small, fully specified set of clinical laboratory questions can be answered reliably and transparently by a generate → execute → validate → correct → present pipeline built on a single-source-of-truth question registry, and that the same pattern can be extended, cautiously, toward more open-ended LLM-authored queries as long as execution is sandboxed and every result still passes through a deterministic gate before it is trusted. Across 12 supported questions and 159 automated tests (94% line coverage), the deterministic path demonstrated consistent and reliable behaviour across all evaluated registry questions, while the optional Advanced mode demonstrated that the same pattern can be extended to more open-ended queries—at the cost of a weaker, generic validator and a narrower set of expressible code.

### 5.2 Advantages

- Reliable and reproducible by default: the trusted path never calls an external model and is fully deterministic and test-covered.
- Transparent and auditable: every run shows the generated code, the validation outcome, and the full correction history, not just the final answer.
- Maintainable: the config-driven registry keeps the UI, generator, validator, and pipeline in sync; adding a question is a one-place change.
- Safe-by-default extensibility: free-form, LLM-authored code is possible but strictly opt-in and sandboxed, never the default execution path.
- Usable for two distinct audiences: the role-based Doctor / Researcher workspace surfaces the subset of questions and framing relevant to each user without hiding the shared, trusted pipeline underneath.

### 5.3 Disadvantages and limitations

- Restricted, by design, to a fixed set of 12 predefined question types; it cannot answer an arbitrary clinical question without either extending the registry or switching to the weaker Advanced mode.
- Operates on a single static, local CSV rather than a live SQL backend, so it does not reflect data that changes after the file is prepared.

- The rule-based corrector only repairs a small, explicit set of known function-name typos; it cannot fix arbitrary bugs, unlike a learned, fault-aware editor such as Self-Edit (section 7).
- The AST allow-list in the experimental mode blocks loops, imports, and several pandas methods, so it will reject some valid, more complex analyses along with unsafe ones.
- No authentication or multi-user state; all valid-but-empty results are reported uniformly as “no records” without deeper diagnostics.

## 5.4 Limitations of the evaluation

- Evaluation is on one dataset (a MIMIC-III derived merge) and one clinical domain (lab events), so generalization to other tables or hospitals is untested.
- There is no external baseline system or a formal comparison against clinicians' own ad-hoc queries; the “baselines” used in section 3.5 are internal ablations of the pipeline's own guardrails.
- The experimental LLM code-generation mode was exercised in automated tests mostly with a fake/mock LLM to test control flow, rather than benchmarked at scale against real models the way AgentCoder or Self-Edit benchmark theirs.

## 5.5 Future work

- Broaden the question registry, or pair a more capable LLM-assisted router with the same validation guarantees, so more clinical questions can be answered through the trusted path rather than the weaker Advanced mode.
- Replace the static CSV with a live SQL backend so results reflect current data and larger cohorts can be queried efficiently.
- Strengthen the guarded LLM code-generation path: a broader, still-safe AST allow-list; benchmarking across multiple LLM providers; and a learned, fault-aware correction step in the spirit of Self-Edit instead of only rule-based typo fixing.
- Run a small user study with clinicians and researchers comparing the role-based workspace against ad-hoc querying, to validate the usability claims with real users rather than internal review.
- Add authentication and multi-user session state if the system moves beyond a single-user course demo.

# 6. User Manual

## 6.1 System requirements and dependencies

- Python 3.10 or later (developed and tested on Python 3.13).
- The uv package manager (<https://docs.astral.sh/uv/>), used to install and pin all dependencies from pyproject.toml / uv.lock.
- Core dependencies (installed automatically by uv sync): pandas >= 2.2, streamlit >= 1.36, httpx >= 0.27, python-dotenv >= 1.0.
- Development-only dependencies: pytest, pytest-cov, playwright.
- Optional: an API key for Anthropic, OpenAI, or Google Gemini, only if the optional LLM-assisted routing or Advanced AI-writes-code mode is desired. The system is fully functional with no key at all.



- The cleaned dataset `cleaned_merged_dataset.csv` (~130 MB) — downloaded automatically on first run, or placed manually in the project root next to `app.py`.

## 6.2 Installation and running the system

1. Install `uv`, then from the project root run: `uv sync` — this creates a local virtual environment and installs every pinned dependency.
2. (Optional) copy `.env.example` to `.env` and fill in one provider API key to enable LLM-assisted routing and the Advanced code-generation mode; leave it empty to run in fully rule-based / deterministic mode.
3. Ensure `cleaned_merged_dataset.csv` is present in the project root, or simply start the app and let it download the file automatically on first launch.
4. Launch the interactive UI: `uv run streamlit run app.py`, then open the printed local URL (typically `http://localhost:8501`).
5. Alternatively, run a single question end-to-end from the command line: `uv run python main_pipeline.py`.
6. Run the automated test suite (159 tests, coverage report): `uv run pytest`.

## 6.3 Required inputs, by question

| Question                                                                                          | Required inputs                                                         |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 1, 4, 7, 9 — abnormal results / all tests / counts / distinct abnormal tests for an admission     | Admission ID (HADM_ID)                                                  |
| 2, 3, 5, 6, 8 — latest value / trend / first-vs-last / summary stats / abnormal results for a lab | Admission ID + lab test (name or ITEMID)                                |
| 10 — values within a date range                                                                   | Admission ID + lab test + start/end date                                |
| 11 — all admissions for a patient                                                                 | Patient ID (SUBJECT_ID)                                                 |
| 12 — abnormal vs normal counts across all admissions                                              | None (runs across the whole dataset)                                    |
| Natural-language box (any question above)                                                         | A free-text sentence, e.g. “Show hematocrit trend for admission 107521” |

## 6.4 Operation steps

### Automatic, on startup

- The dataset is loaded once (`data_prep.py`) and, if missing, downloaded from the configured source.
- The registry-driven function context used by the code generator/executor is built from `questions.py`.

### Manual, per query

1. From the Welcome screen, choose the Doctor or Researcher workspace (or use the direct question dropdown).
2. Pick a category and a question, or expand “Ask a clinical lab question in natural language” and type one.
3. Fill in the parameter widgets that appear for the chosen question (only the parameters that question needs are shown).

4. Optional: choose an interpretation mode for natural-language input (Rules only / Rules, then AI / Advanced: AI writes code), or tick the “inject a typo” demo checkbox to see the correction loop fire.
5. Click Run Analysis (or Interpret & Run for the natural-language box).

## 6.5 Outputs

- A colour-coded status line (green = passed, blue = no matching records, red = failed).
- Summary metric cards appropriate to the question (e.g. latest value, first-vs-last difference, count/min/max/mean, abnormal counts).
- A trend line chart for time-series questions (trend and date-range questions).
- Final Result tab: the result table with a “Download Result as CSV” button.
- Validation Audit tab: which structural/clinical rule was checked and whether it passed.
- Correction Attempts tab: every attempt’s code, execution outcome, and validation outcome (“No correction attempts were needed” in the normal case).
- Generated Code tab: the exact Python snippet that was executed to produce the result (for the Advanced mode: Final Result / AI Process / Generated Code / Attempts tabs).

## 6.6 Repository / code structure

| Path                                                                | Contents                                                                      |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------|
| app.py                                                              | Streamlit UI                                                                  |
| main_pipeline.py                                                    | End-to-end console pipeline                                                   |
| questions.py                                                        | Question registry (single source of truth)                                    |
| question_router.py, llm_router.py, llm_client.py                    | Controlled / optional LLM-assisted natural-language routing                   |
| data_prep.py                                                        | Dataset load + per-question data functions                                    |
| code_generation.py                                                  | Template code generation                                                      |
| execution.py                                                        | Registry-path execution wrapper                                               |
| validation.py                                                       | Output validation                                                             |
| correction.py                                                       | Rule-based correction loop                                                    |
| result_presentation.py                                              | Console presentation                                                          |
| sandbox.py, code_agent.py, agent_pipeline.py, presentation_agent.py | Experimental, sandboxed LLM code-generation mode                              |
| tests/                                                              | 159 unit + end-to-end tests                                                   |
| docs/                                                               | PRD, PLAN, TODO, REPORT, DEMO, SLIDES, TEST_REPORT, decision log, screenshots |
| pyproject.toml, uv.lock                                             | Dependency management (uv)                                                    |

## 7. Articles Read

### 7.1 Abstracts

#### **AgentCoder: Multi-Agent-based Code Generation with Iterative Testing and Optimisation**

Huang, D., Zhang, J. M., Luck, M., Bu, Q., Qing, Y., & Cui, H. (2023, revised 2024). *arXiv:2312.13010*.

<https://arxiv.org/pdf/2312.13010>

The paper addresses a persistent weakness of LLM-based code generation: a single model asked to both write and check its own code tends to balance these two jobs poorly, generating code that looks plausible but fails on real test cases. The authors propose AgentCoder, a multi-agent framework with three specialized, cooperating roles: a programmer agent that writes and refines code, a test designer agent that independently generates test cases for that code, and a test executor agent that runs the code against those tests and reports concrete feedback (pass/fail, error messages) back to the programmer. This separation of concerns — write, independently test, execute and report — lets the programmer iterate using real execution feedback instead of the model's own untrustworthy self-assessment. The main contribution is showing that this lightweight three-agent loop beats both single-agent prompting and heavier multi-agent frameworks: across 9 code-generation models and 12 enhancement approaches, AgentCoder with GPT-4 reaches 96.3% and 91.8% pass@1 on HumanEval and MBPP respectively, versus 90.2%/78.9% for the prior state of the art, while using roughly a third of the token overhead — evidence that more agents/rounds are not automatically better, and that a small, well-separated set of roles with real execution feedback is what actually drives the improvement.

#### **Self-Edit: Fault-Aware Code Editor for Code Generation**

Zhang, K., Li, Z., Li, J., Li, G., & Jin, Z. (2023). *Proceedings of ACL 2023*.

<https://aclanthology.org/2023.acl-long.45v3.pdf>

This paper starts from the same observation that motivates AgentCoder — LLMs generate code with real but limited accuracy on competitive-programming-style tasks — but proposes a lighter, generate-and-edit pipeline instead of a multi-agent one. Self-Edit has three steps: an LLM generates an initial program from the problem description; the program is executed on the example test case(s) given with the problem, and any error or wrong output is wrapped into a structured “supplementary comment” (e.g. the exact exception and a hint at the fix); and a separate, purpose-trained fault-aware code editor model then rewrites the code conditioned on the original program, the problem description, and that comment. Evaluated across 9 LLMs (110M–175B parameters) on the APPS and HumanEval benchmarks, Self-Edit improves average pass@1 by 89% on APPS-dev, 31% on APPS-test, and 48% on HumanEval relative to generating directly from the LLM, and does so with a constant, small sampling budget — outperforming reranking-based post-processing methods that instead sample many candidates and pick one. The key contribution is demonstrating that execution feedback, distilled into a compact structured comment, is enough signal for a dedicated editor model to reliably repair a large fraction of an LLM's own coding mistakes.

### 7.2 Contribution of each article to the project

#### **AgentCoder → the optional Advanced code-generation mode**

AgentCoder's generate → test → execute → feedback loop, and specifically its finding that a small set of clearly separated roles beats a heavier multi-agent framework, influenced the design of the project's

experimental “Advanced: AI writes code” path. `agent_pipeline.py` implements the same generate → execute → validate → refine shape: `code_agent.py` plays the programmer-agent role, `sandbox.py` plays the (deterministic, non-LLM) executor role, and `validation.validate_freeform_result` plays a lightweight stand-in for the independent test/validator role. Following AgentCoder’s own critique of frameworks such as MetaGPT or ChatDev — that stacking more agents mainly adds token and coordination overhead once the necessary roles exist — the project deliberately kept this orchestration lightweight and in-repo rather than adopting CrewAI, AutoGen, or LangGraph, reusing two roles (execution, validation) that the project’s own deterministic modules already provided.

### Self-Edit → the rule-based correction loop

Self-Edit’s core idea — run the generated code, turn the execution result into structured feedback, and use that feedback to edit the code before resubmitting — served as the conceptual model for `correction.py`, whose own module docstring describes it as “Correction module (a simplified, rule-based interpretation of the Self-Edit approach)”. The project’s corrector is a deliberately simplified, rule-based analogue of Self-Edit’s learned fault-aware editor: instead of a trained neural editor conditioned on an execution comment, `correct_generated_code()` applies a small, explicit table of known function-name-typo fixes to the code and re-runs it, still following Self-Edit’s generate → execute → (structured feedback) → edit → resubmit shape, but with a deterministic rule set in place of a learned model — appropriate for a bounded, fully-specified question registry where the space of likely code-generation faults is small and known in advance. The paper’s broader finding, that execution feedback alone (without re-sampling from the LLM) is enough signal to repair a meaningful share of generation errors, is the justification for keeping correction rule-based and cheap rather than requiring another model call every time execution fails.